

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Week No.: 2

Student Name: B.Harsha Vardhan

Roll No.: A24126552075

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1. TITLE

Design Webpage Using HTML5 & CSS3

2. AIM

To construct a styled webpage using HTML5 semantic elements (, , ,) and CSS3 layout styles.

3. OBJECTIVE

The objectives of this experiment are:

- Implement HTML5 semantic sectioning elements.
- Apply CSS3 linear gradient background and navigation styling.
- Construct a 2-column Main Content and Sidebar layout.

4. THEORY

HTML5 semantic tags provide meaningful document structure. CSS3 styling rules enable linear background gradients, flexible 2-column flexbox layouts, callout highlight boxes, and styled sidebar widgets.

5. ALGORITHM / PROCEDURE

1. Create index.html and style.css files.
2. Set up header banner with dark gradient background.
3. Build 2-column flexbox container with main article and sidebar.
4. Style callout highlight box and topic tags.
5. Verify responsive layout in browser.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Tech Insights Journal</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

<header>
```

```
<h1>Tech Insights Journal</h1>
<p>Exploring Modern Web Technologies & Artificial Intelligence</p>
</header>

<nav>
  <a href="#">Articles</a>
  <a href="#">Tutorials</a>
  <a href="#">Resources</a>
  <a href="#">About Author</a>
</nav>

<div class="layout-wrapper">

  <article class="main-article">
    <h2>The Future of Full Stack Web Development</h2>
    <p class="meta-info">Published on August 30, 2026 | Category: Web Architecture</p>

    <p>Modern web development has evolved rapidly with semantic HTML5 standards and CSS3 layout systems. Developers can now construct high-performance web applications with clean structure and responsive styling.</p>

    <div class="highlight-box">
      <h3>Key Takeaway</h3>
      <p>Combining HTML5 semantic tags with CSS3 flexbox and grid layouts ensures accessible, responsive, and maintainable web applications.</p>
    </div>

    <p>Using CSS3 linear gradients, custom shadows, and media queries, web designers can create immersive user experiences without relying on heavy external graphics.</p>
  </article>

  <aside class="sidebar">
    <div class="profile-widget">
      <h3>Author Profile</h3>
      <p><b>Name:</b> B.Harsha Vardhan</p>
      <p><b>Roll No:</b> A24126552075</p>
      <p><b>Course:</b> Full Stack Web Dev (23CM4121)</p>
    </div>

    <div class="tags-widget">
      <h3>Popular Topics</h3>
      <span class="tag">HTML5</span>
      <span class="tag">CSS3</span>
      <span class="tag">JavaScript</span>
      <span class="tag">AI & ML</span>
    </div>
  </aside>

</div>

<footer>
  <p>© 2026 B.Harsha Vardhan | Full Stack Development Lab</p>
</footer>

</body>
</html>
```

style.css

```
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  font-family: Arial, sans-serif;
  background: #eef2f6;
  color: #222;
}

header {
  background: linear-gradient(135deg, #1e293b, #0f172a);
  color: white;
  text-align: center;
  padding: 30px 15px;
}

header h1 {
  font-size: 26px;
  margin-bottom: 6px;
}

header p {
  font-size: 14px;
  color: #94a3b8;
}

nav {
  background: #0284c7;
  text-align: center;
  padding: 12px;
}

nav a {
  color: white;
  text-decoration: none;
  margin: 0 15px;
  font-size: 15px;
  font-weight: bold;
}

nav a:hover {
  color: #e0f2fe;
}

.layout-wrapper {
  display: flex;
  max-width: 1000px;
  margin: 30px auto;
  gap: 20px;
  padding: 0 15px;
}

.main-article {
  flex: 2;
  background: white;
  padding: 25px;
}
```

```
    border-radius: 8px;
    box-shadow: 0 3px 10px rgba(0,0,0,0.06);
}

.main-article h2 {
    color: #0f172a;
    font-size: 22px;
    margin-bottom: 8px;
}

.meta-info {
    font-size: 12px;
    color: #64748b;
    margin-bottom: 15px;
}

.main-article p {
    font-size: 14px;
    line-height: 1.6;
    color: #334155;
    margin-bottom: 15px;
}

.highlight-box {
    background: #f0f9ff;
    border-left: 4px solid #0284c7;
    padding: 15px;
    margin: 20px 0;
    border-radius: 4px;
}

.highlight-box h3 {
    color: #0369a1;
    font-size: 15px;
    margin-bottom: 5px;
}

.highlight-box p {
    margin-bottom: 0;
    color: #0c4a6e;
}

.sidebar {
    flex: 1;
    display: flex;
    flex-direction: column;
    gap: 20px;
}

.profile-widget, .tags-widget {
    background: white;
    padding: 20px;
    border-radius: 8px;
    box-shadow: 0 3px 10px rgba(0,0,0,0.06);
}

.sidebar h3 {
    font-size: 16px;
```

```
    color: #0f172a;
    margin-bottom: 12px;
    border-bottom: 2px solid #e2e8f0;
    padding-bottom: 6px;
}

.profile-widget p {
    font-size: 13px;
    margin-bottom: 8px;
    color: #475569;
}

.tag {
    display: inline-block;
    background: #e0f2fe;
    color: #0369a1;
    padding: 5px 10px;
    border-radius: 15px;
    font-size: 12px;
    font-weight: bold;
    margin: 4px 2px;
}

footer {
    background: #0f172a;
    color: #94a3b8;
    text-align: center;
    padding: 15px;
    margin-top: 40px;
    font-size: 13px;
}
```

7. CODE EXPLANATION

Code / Syntax	Explanation
<header>, <nav>, <article>, <aside>	HTML5 semantic page containers
background: linear-gradient(...)	Dark gradient background for header banner
display: flex; gap: 20px;	Flexible 2-column main content and sidebar layout
.highlight-box	Callout box with accent left border
.tag	Rounded pill tags for topic keywords

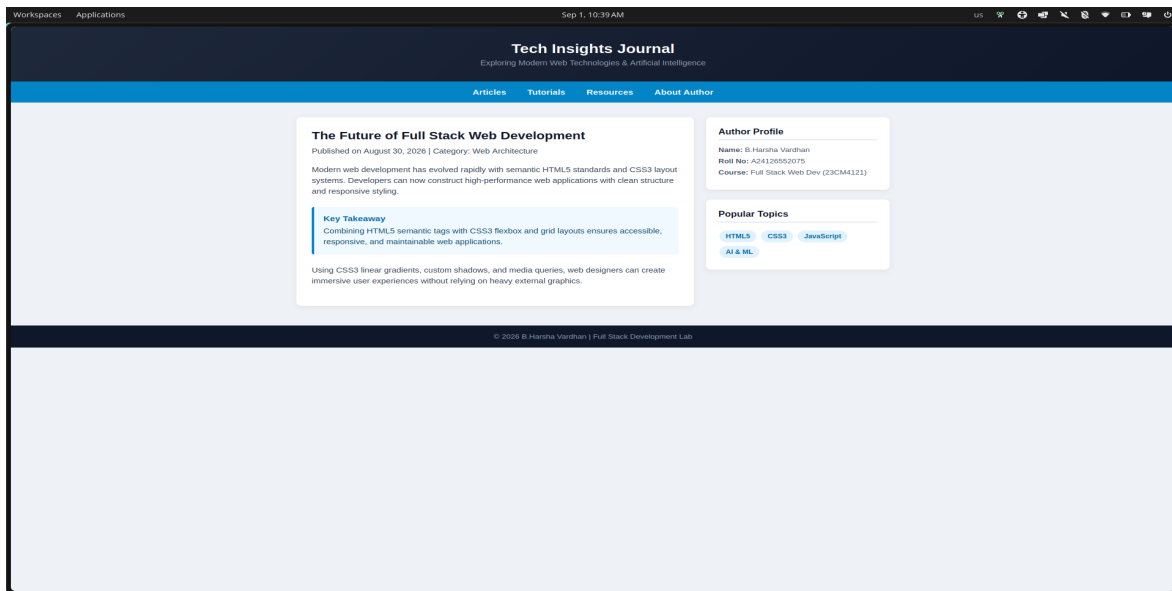
8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Header & Nav	Render header banner and nav links	Render header banner and nav links	Pass
TC-02	2-Column Layout	Main article and sidebar align horizontally	Main article and sidebar align horizontally	Pass
TC-03	CSS Callout Box	Highlight box displays with accent border	Highlight box displays with accent border	Pass

9. OUTPUT

Output 1: Execution Screenshot



10. RESULT

Thus, the experiment for Week 2 (Design Webpage Using HTML5 & CSS3) was successfully executed and verified.